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RESEARCH ARTICLE

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Improved search and rescue optimization based load balancing scheme of CCTV footage on edge computing environment

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Summary

Nowadays, data centers and servers are rising in size by order of magnitude due to increasing CCTV footages. Therefore, load balancing (LB) becomes a primary concern in scalable computing system involves mobile edge cloud computing environment. In mobile edge cloud computing system, a mobile user offloading their task to neighboring edge server for supporting realistic application. But while user is situated in a hot spot, various edge servers are over-loaded as a result of sudden offloading task from the mobile user. The LB process can be considered as an optimization problem, which can be resolved by the use of metaheuristic optimization algorithms. This article introduces an Improved Search and Rescue Optimization based LB Scheme for Edge Computing Environment (ISROLBS-ECE). The presented ISROLBS-ECE technique majorly concentrates on the distribution of offloading tasks to neighboring edge servers effectively. In addition, the ISROLBS-ECE technique involves the hybridization of chaotic concept into the conventional search and rescue optimization (SRO) approach. In the ISROLBS-ECE technique, the moderately lightest and task arrival nodes are employed as target node for allocating new task whereas rest of the nodes are not provisionally allocated tasks to accomplish dynamic balancing of the network. The experimental validation of the ISROLBS-ECE technique is tested and the outcomes are investigated under various aspects. The comparative study shows the better performance of the ISROLBS-ECE approach over other recent techniques.

KEYWORDS

CCTV footage, edge computing, load balancing, search and rescue optimization, task allocation, task offloading

1 | INTRODUCTION

In recent times, mobile devices (like computers, smart phones, tablet, and so on) are very helpful tools for updating news and businesses, learning, social networking, entertainment. Initially, mobile users are not having same fulfillment when comparison made with desktop because of resource limits of mobile gadgets.[1] With the incredible progressions in networking and wireless communications, Mobile Cloud Computing (MCC) becomes an emergent as an innovative pattern of computing for the past few years for improving mentioned scenario. With MCC, the higher rate and highly dependable air interface permits resource limitations user gadgets to offload certain computation to remote resourceful cloud. Once the emergence of MCC, several CC services like mobile gaming, mobile health care, and mobile learning is straight accessible from mobile gadgets.[2,3]

With the arrival of CC, the back-end server can be commonly hosted at the cloud data center. Even though the usage of a cloud data center provides numerous advantages namely elasticity and scalability, its centralization and consolidation results in a huge separation among a mobile

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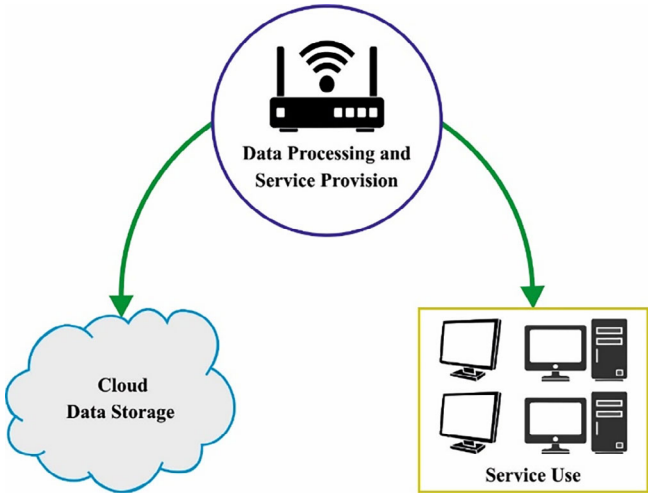


FIGURE 1 General edge computing environment

gadget and its related data center. Offload computation to public cloud might include low bandwidth and long latency for data interchange among the edge device and public clouds via the Internet.[4] Therefore, MCC was not sufficient for an extensive range of evolving mobile applications, which were latency critical. Mobile Edge Computing (MEC) "presents IT and CC abilities in the Radio Access Network (RAN) in closer contact to mobile users." In MEC, the duty of edge can be raised. Service and computation were permitted in cloudlets at the edge that invokes numerous proximal MEC servers. Mobile application merit from MEC by offloading its computationally intensive errands to the MEC server, focusing to minimalize the bandwidth consumption and network latency.[5] There were three fundamental elements in the method of MEC one is mobile devices has all kinds of gadgets (both IoT gadgets and cell phones) linked to the network; next MEC servers were commonly small-scale data centers positioned in close contact with users. It is responsible for conventional network traffic control (filtering and forwarding) and hosting several mobile edge applications (smart tracking, edge health care, and much more) and lastly public cloud becomes cloud structure hosted in the Internet.[6] Figure 1 shows the general structure of edge computing environment.

Though edge computing could importantly minimize latency, the irrational task assignment results in the unbalancing load of every node.[7] And due to the heterogeneity and diversity of edge computing sensors, the common load balancing (LB) method may not be strictly implemented to edge computing. Thus, LB and edge computing are the most significant research subject in academia. There exist two major kinds of LB policies they are dynamic and static.[8] LB policy by task allotment in edge computing depends upon intermediary node was a dynamic LB approach that takes the former state of a node at the time of load distribution.[9,10] Static LB methods not considered the earlier state of node, whereas dispensing the load and it functions well whenever node contains a slight difference in the load. As a result, it does not suit the edge atmosphere. In this study, method is considered for reducing the processing duration of these applications, when assuming the edge network features. The fact that several machine learning (ML) and data analytics techniques is written in a "summation form", allows to speed up by making use of distributed computing sources. In this study, efforts are made for optimizing speed up, considering the edge network limitations. In particular, the formulation of a LB optimization issue to execute a learning technique on edge devices in an edge network.

The main contributions of this paper are as follows:

- In the MEC environment, an ISROLBS-ECE technique-based task offloading and resource allocation algorithm is proposed. The algorithm learns the strategy for selecting suitable computing nodes in an adaptive manner.
- The proposed algorithm optimises the task average response time and total system energy consumption based on the profits of users and service providers.
- The mobility of User Equipment (UE) is considered to make the experimental environment more realistic. The results of our experiments show that our proposed algorithm performs well in terms of reducing average response time and total energy consumption.

This paper presents an Improved Search and Rescue Optimization based Load Balancing Scheme for Edge Computing Environment (ISROLBS-ECE). The goal of the ISROLBS-ECE technique is to distribute offload task to neighboring edge servers effectively. Besides, the ISROLBS-ECE technique involves the hybridization of chaotic concept into the conventional search and rescue optimization (SRO) technique. This study proposes an SRO adaptive model based on the characteristics of edge cloud task requests and the Edge Computing environment. The fitness and load balancing results of the SRO model were obtained by designing increased solution space scenarios in conjunction with chaotic sequence

generation theory. Experiments show that in the simulation experiment, the fitness of this model is 22% higher than that of the comparison algorithm, and the model still maintains the minimum standard deviation in the cluster single-index comparison. In the ISROLBS-ECE algorithm, the comparatively lightest and task arrival nodes are employed as target nodes for allocating new task whereas rest of the nodes are not provisionally allocated task to accomplish dynamic balancing of the network. The performance evaluation of the ISROLBS-ECE technique is validated and the results are examined under varying aspects.

2 | LITERATURE REVIEW

In Reference 11, an energy aware method based on the marine predator algorithm (MPA) was suggested to tackle the task scheduling in fog computing (TSFC) for improving the QoS required by end-users. Along with the typical MPA, the additional two versions also suggested. The primary version termed modified MPA (MMPA), that makes a modification in MPA for improving its exploiting ability with the help of the last upgraded positions rather than the last optimal one. The next one would enhance MMPA by ranking strategies related mutation and reinitialization to the optimum, along with the reinitializing, the half population arbitrarily get rid of local optima after a predefined quantity of iterations and mutate the final half against the best-so-far solution. Dai et al.[12] suggest integrating LB to that of offloading and learn resource allotment for a multi-user multi-server VEC mechanism. First, the offload problem and joint LB were expressed as a mixed integer nonlinear programming issue for maximizing utilities of system. Mainly, IEEE 802.11p protocol is considered to model the utilities of system. After, decoupling the issue as two subtasks and advance a lower complexity method to mutually make VEC server choice, and enhance computational resource and offloaded ratio.

In Reference 13, a multiple UAV-enabled mobile-edge computing (MEC) mechanism can be built, whereas many UAVs serves as MEC nodes for providing computed offloaded service for ground IoT sensor nodes that contain constrained local computed abilities. For load balancing of UAV, the differential evolution (DE)-related multiple UAV deployment system was suggested, wherein the accessing issue is modeled as a generalized assignment problem (GAP), that can be after resolved by a near-optimal solution method. Zhang et al.[14] launch fiber-wireless (FiWi) technology for enhancing VECNs, because of its benefits on central network management and support many transmission methods. Aims at reducing the process delay of the computational tasks of vehicles, software defined networking (SDN) related LB task offloaded method was recommended in WiFi advanced VECN, whereby SDN can be launched for offering support for the vehicle information management and centralized network.

In Reference 15, two nature-inspired metaheuristic schedulers, like PSO, and ant colony optimization (ACO) were utilized for suggesting two distinct scheduling techniques to balance effectually load IoT errands on fog nodes in response time considerations and transmission cost. In Reference 16, a blockchain (BC)-enabled computation offloaded technique, called BeCome, was suggested in this article. BC can be utilized in edge computing for data integrity. After which, the non-dominated sorting GA III can be applied to produce strategies for balanced source allotment. In Reference 17, deep reinforcement learning (DRL) can be primarily suggested for solving the offload issue of many service nodes for the cluster and many dependencies for mobile errands in largescale heterogeneous MEC. The work utilizes the the candidate network set and LSTM network layer for enhancing the DQN system in amalgamation with the actual atmosphere of the MEC.

Bendechache et al.[18] discussed resource allocation. Resource provisioning and scheduling metrics were explored. Provisioning metrics include detection, selection, and mapping; scheduling metrics include allocation, monitoring, and load-balancing. Scalability, latency, VM placement, failure rates, accuracy, resource utilization, energy consumption, cost, efficiency, SLA, and QoS were also investigated. This research survey is detailed, but it presents few cloud, fog, and edge computing research articles. This motivates the study of cloud, fog, and edge resource management.

In Reference 19, three taxonomies are shown: (i) performance metrics for evaluating cloud, fog, and edge computing; (ii) metrics based on cloud models; and (iii) classification of identified metrics based on MAPE-K. Throughput, network congestion, fault-tolerance, statistical analysis measurements, scalability, cost/profit, and SLA violation are common cloud, fog, and edge computing performance metrics. Private, public, hybrid, single-provider, multi-provider, and federated are cloud model metrics taxonomies. Monitoring, analyzing, planning, and executing are MAPE-K loop parameters. Their results map proposed taxonomy to cloud, fog, and edge computing literature. The study could be expanded by listing proposed solutions from the reviewed literature and classifying them.

Ghobaei-Arani et al.[3] categorized fog computing resource management. The taxonomy includes provisioning, placement, scheduling, offloading, load balancing, and allocation. They organized literature by resource management approaches. They described each resource management approach's case study, technique, performance metric, evaluation tool, advantages, and weaknesses. This study provided knowledge about existing articles for each resource management approach but did not include edge computing. The article only explores solutions. The research examines and discusses existing resource management studies. In Reference 20, IoT appliances in cloud, fog, and edge computing face resource and load balancing challenges. Load balancing is discussed to efficiently use resources and avoid congestion.

Load balancing IoT processing nodes in a fog environment is a unique challenge. According to, the challenge was to reduce latency and workload to save energy.

3 | THE PROPOSED MODEL

In this study, a new ISROLBS-ECE methodology has been developed for distributing the offloaded tasks to neighboring edge servers effectively. Besides, the ISROLBS-ECE technique involves the hybridization of chaotic concepts into the traditional SRO algorithm. The workflow of presented approach is shown in Figure 2. In an Edge computing environment, this model manages the resource allocation. Three layers make up the cloud environment model: a client layer, a fog layer, and a cloud layer. To fulfill the need for resources for clients, a resource management solution is first realized in the client and fog layers. The request is forwarded to the cloud layer if there are no or few resources available in the fog layer. The following are the main functional elements of this model:

- All the processors that are available are used by the client by the fog server manager.
- With the aid of a "server virtualization technique," fog servers are composed of a fog server manager and virtual machines (VMs) that operate requests for the fog data server, process them, and then deliver the results to the manager.
- Fog servers contain one fog server manager and virtual machines to conduct requests by using a "server virtualization technique."

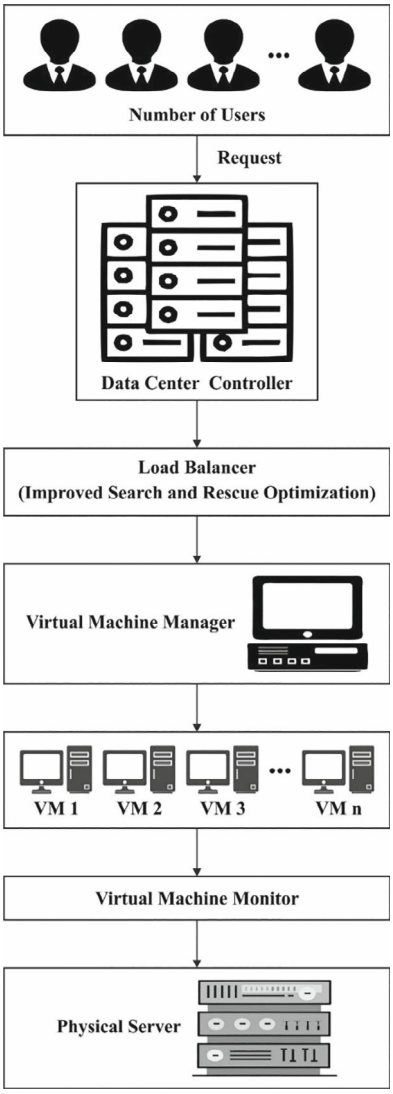


FIGURE 2 Process in proposed load balancing model

3.1 | Design of ISRO algorithm

In SRO methodology, the position of human is corresponding to the solution of optimization issue, also the quantity of clues accomplish in this position characterizes the objective function for that solution. The group member gathers clue information in the search. Few clues are left in the event of accomplishing best possible clue in another location, on the other hand, the dataset on them are applied to enrich the search technique.[21] In the study, the left clue location is saved in the memory matrix M, whereby the location of individual is stored in the X matrixes. The dimension of M matrix is comparable to X matrix. They are $N \times D$ matrices, where D indicates problem dimension and N specifies amount of humans. The clue matrixes comprises the position of accomplished clues. The matrix, M, and C are upgraded in human searching phase:

$$C = \begin{Bmatrix} X \\ M \end{Bmatrix} = \begin{Bmatrix} X_{11} & \cdots & X_{1D} \\ \vdots & \ddots & \vdots \\ X_{N1} & \cdots & X_{ND} \\ M_{11} & \cdots & M_{1D} \\ \vdots & \ddots & \vdots \\ M_{N1} & \cdots & M_{ND} \end{Bmatrix}. \tag{1}$$

From the equation, M and X indicates memory and human location matrix, and X_{N1} signifies position of primary dimension for Nth human. Likewise, M_{1D} characterizes location of Dth dimensional vector for the primary memory. Consider the description given in the earlier subsection, besides a random clues among accomplished clues, the search direction is described below:

$$\mathrm{SD}_i = (X_i - C_k), k \neq i, \tag{2}$$

In Equation (2), X_i, C_k, and SD_i indicates jth human location, kth clue location, as well as search direction of jth human. k indicates arbitrary number lies within 1 and $2N$ and chosen in a way $k \neq i$. Generally, it is important to highlight that human try to find in the way that each desirable area is searched and few repetition location could not be searched again. Therefore, the search should be generated whereby the motion of group member to one another is constraint. TTo apply these limitations, the binomial crossover operator was applied. Once the provided clue is high in contrast with the clue interconnected with the existing position, a nearby region SD_j direction and the position of searched clues; next, the search process undergoes nearby the present location besides the SD_i direction:

$$X'_{i,j} = \begin{cases} \begin{cases} C_{k,j} + rl \times (X_{i,j} - C_{k,j}), & \text{if } f(C_k) >, f(X_i) \\ X_{i,j} + rl \times (X_{i,j} - C_{k,j}), & \text{otherwise} \end{cases} & \text{if } r2 < \mathrm{SE} \text{ or } j = j_{\mathrm{rand}}, \\ X_{i,j} & \text{otherwise}, \end{cases} \quad (j = 1, \ldots, D), \tag{3}$$

In Equation (3), $X'_{i,j}$ symbolizes novel locationof jth parameter of ith human; $C_{k,j}$ denotes the location of jth parameter for the kth accomplished clue; $f(C_k)$ and $f(X_i)$ correspondingly indicates the objective function for solution C_k and X_i; rl and $r2$ indicates random numbers; j_{rand} represent arbitrary value ranges from 1 and D that assurance that as a minimal of 1D of $X'_{i,j}$ is dissimilar from $X_{i,j}$; and Equation (3) is employed for accomplishing an original position of jth humans in all the dimensions.

Next, human searches near the existing location, also the idea of relating dissimilar clues are applied in the social phases. Conflict to the social phase, all the dimensions of X_i attuned in the separation phase. The novel location of jth human can be accomplished as:

$$X'_i = X_j + r3 \times (C_k - C_m), i \neq k \neq m, \tag{4}$$

In Equation (4), k and m represents random value ranges from 1 and $2N$. To prevent motion besides other clues, k and m are elected in this way that $i \neq k \neq m$. $r3$ represent an arbitrary number using standard distribution ranges from [0, 1]. In each metaheuristic approach, every solution need to be situated in the solution space, and once they are far from the solution space, they should be modified as follows[22]:

$$X'_{i,j} = \begin{cases} \frac{\left(X_{i,j} + X_j^{\max}\right)}{2}, & \text{if } X'_{i,j} > x_j^{\max}, \\ \frac{\left(X_{i,j} + X_j^{\min}\right)}{2}, & \text{if } X'_{i,j} < x_j^{\min}, \end{cases} \quad (j = 1, \ldots, D). \tag{5}$$

Following, in all the iterations, the member of group search on the basis of the two processes, and afterward every phase, once the value of objective function in location $X_i'\left(f\left(X_i'\right)\right)$ is high than preceding one ($f\ (X_i)$), the earlier location (X_i) is saved in an arbitrary location of M memory matrix and the position is adopted as a novel position as follows. Next, these locations are left and memory is not upgraded:

$$M_n = \begin{cases} X_{\mathrm{i}}, \text{if } f\left(X_i'\right) >, f\left(X_i\right) \\ M_n, \text{or else} \end{cases}, \tag{6}$$

$$X_i = \begin{cases} X_{\mathrm{i}}', \text{if } f\left(X_i'\right) >, f\left(X_i\right) \\ X_i, \text{or else}, \end{cases} \tag{7}$$

From the expression, M_n signifies location of nth clue stored in the memory matrixes and n signifies arbitrary value ranging within 1 and N. Using these kinds of memory, upgrading rises the dissimilar kinds of technique and the capacity of method to distinguish a global optimum solution. Initially, USN is set as 0 for all the humans. As soon as the human detect a best possible clue in the initial or following phases of the search technique, the USN is set as zero for that human; or else, it raises by one point as follows[23,24]:

$$\mathrm{USN}_i = \begin{cases} \mathrm{USN}_i + \mathrm{I}, \text{if } f\left(X_i'\right) <, f\left(X_j\right) \\ 0, \text{or else} \end{cases}. \tag{8}$$

The arbitrary position in the searching region is represented in the following, and USN_i is set as zero for that human:

$$X_{i,j} = X_j^{\min} + r4 \times \left(X_j^{\max} - X_j^{\min}\right)\ j = 1, \ldots, \mathrm{D}. \tag{9}$$

In Equation (9), $r4$ represents a random value and is diverse for each dimension.

The ISRO algorithm is derived by the use of chaotic maps for balancing the trade-off among exploration and exploitation stages. A chaos map makes use of chaotic variables with erratic nature rather than arbitrary variables. Here, the circle chaotic map function is applied to increase the convergence rate of the SRO algorithm and discover the searching area effectively to attain improved performance. To increase the effectiveness of the SRO approach, the arbitrary numbers gets substituted with the random number of Chaos function:

$$K_{j+1} = \mathrm{mod}\ \left(k_j + b - \frac{a}{2\pi}\sin\left(2\pi K_j\right), 1\right). \tag{10}$$

In Equation (10), K_{j+1} denotes chaotic arbitrary number produced at present round, K_j defines chaotic random number produced in the preceding iteration. a and b are equivalent to 0.5 and 0.2, correspondingly, and we assume $k0$ equal to 0.01 (Algorithm 1).

Algorithm 1. Pseudo-code of the SRO algorithm

Start:

Initialize parameter and uniform distribution population randomly $\left[X_j^{\min}, X_j^{\max}\right], j = 1, \ldots, D$

Arrange the solution in a decreasing manner and define the optimum location (Xbest)

Apply the initial half of the solution for X and remaining for M

Characterize the algorithmic variables (SE, MU) and set $\mathrm{USN}_i = 0$ where $i = 1, \ldots, N$

While ending criteria is not satisfied do

For $i = 1$ to N do

Social phase

$$C = \left\{ \begin{matrix} X \\ M \end{matrix} \right\}$$

$\mathrm{SD}_j = (X_j - C_k)$, k is arbitrarily chosen in this way that $i \neq k$

$j_{\mathrm{rand}} = \mathrm{rand\ int}\ [1, D]$

$r1 = \mathrm{rand}\ [-1, 1]$

For $j = 1$ to D do

$$X_{i,j}' = \begin{cases} C_{k,j} + r1 \times \mathrm{SD}_{i,j}, & \text{if } f\left(C_k\right) > f\left(X_i\right), \\ X_{i,j} + r1 \times \mathrm{SD}_{i,j}, & \text{or else} \end{cases}$$

$$X_{i,j}' = \begin{cases} \left(X_{i,j} + X_j^{\max}\right)/2, & \text{if } X_{i,j}' > X_j^{\max} \\ \left(X_{i,j} + X_j^{\min}\right)/2, & \text{if } X_{i,j}' < X_j^{\min} \end{cases}$$

End For

$$M_n = \begin{cases} X_i, & \text{if } f\left(X_i'\right) > f\left(X_i\right), \\ M_n, & \text{or else} \end{cases}$$

$$X_i = \begin{cases} X_i', \text{if } f\left(X_i'\right) > f\left(X_i\right) \\ X_i, \text{or else} \end{cases}$$

$$\text{USN}_i = \begin{cases} \text{USN}_i + 1, \text{if } f\left(X_i'\right) < f\left(X_i\right) \\ 0, \text{or else} \end{cases}$$

Individual phase

$$C = \begin{cases} X \\ M \end{cases}$$

$X_i' = X_i + \text{rand}\,[0, 1] \times (C_k - C_m)$, k and m are arbitrarily chosen in this way that $i \neq k \neq m$

For $j =$ l to D do

$$X_{i,j}' = \begin{cases} \left(X_{i,j} + X_j^{\max}\right)/2, & \text{if } X_{i,j}' > X_j^{\max} \\ \left(X_{i,j} + X_j^{\min}\right)/2, & \text{if } X_{i,j}' > X_j^{\min} \end{cases}$$

End For

$$M_n = \begin{cases} X_i, & \text{if } f\left(X_i'\right) > f\left(X_i\right), \\ M_n, & \text{or else} \end{cases} \; n \text{ randomly chosen}$$

$$X_i = \begin{cases} X_i \text{ if } f\left(X_j'\right) > f\left(X_i\right) \\ X_i \text{ or else} \end{cases}$$

$$\text{USN}_i = \begin{cases} \text{USN}_i + 1, \text{if } f\left(X_i'\right) < f\left(X_i\right) \\ 0, \text{or else} \end{cases}$$

If $\text{USN}_i >$ MU do

For $j =$ l to D do

$X_{i,j} = X_j^{\min} + \text{rand}\,[0, 1] \times \left(X_j^{\max} - X_j^{\min}\right)$

End For

$\text{USN}_i = 0$

End If

End For

Define the current optimum location and upgrade Xbest

End while

Return Xbest

End

3.2 | Process involved in load balancing scheme

If more than one task arrives at node n_i at the same time, the task is combined into an aggregated task U. Based on the data of the targeted nodes, the aggregated tasks are decomposed into many subtasks $u_j = \alpha_j U$ that is managed by distinct targeted, and task arrived nodes. The transmitting time is the size of subtask separated by the data transfer rate from n_i edge nodes to n_j target node or d cloud server, as follows:

$$t_1 = \frac{\alpha_j U}{B_{n_i, n_j'}}, \tag{11}$$

$$t_2 = \frac{\alpha_d U}{B_{n_i, d}}. \tag{12}$$

Among other, $B_{n_i, n_j'}$, refers to the data transmission rate of n_i edge node to the n_j target node, and $B_{n_i, n_j} = \infty$ if $n_i = n_j$.[3] $B_{n_i, d}$ is the data transmission rate of edge node n_i to the d cloud server.

The computational time of subtask is given as follows.

$$t_3 = \frac{\alpha_j U}{f_{n_j}} + \tau_{n_i, n_j} \frac{N_{nj}}{f_{n_j}}, \tag{13}$$

$$t_4 = \frac{\alpha_d U}{f_d} + \tau_{n_j, d} \frac{N_d}{f_d}, \tag{14}$$

where, $\frac{\alpha_j U}{f_{n_j}}$ refers to the computation time of the subtasks at the targeted node n_j, and f_{n_j} is the calculation speed of targeted node n_j. $\tau_{n_j,n_j}\frac{N_{n_j}}{f_{n_j}}$ indicates the existing task computation time of the targeted node n_j. And τ_{n_i,n_j}, $= \lceil \alpha_j \rceil$, it indicates either there is a task allocation relationships among the edge nodes n_i and n_j. $\tau_{n_i,n_j} = 1$ denotes that the relationships exist; $\tau_{n_i,n_j}, = 0$ represent that the relationships does not takes place. N_{n_j}, indicates the existing task size of target node n_j. Likewise, we get a description of the calculation time t_4 of subtask in cloud server.

In general, the computation outcome is a smaller packet namely a control signals; therefore, the transmitting time of computation outcome is disregarded.[24]

- The execution time of subtasks among edge nodes

$$T_1(\alpha_j) = \max(t_1 + t_3) = \max\left(\frac{\alpha_j U}{B_{n_i,n_j}} + \frac{\alpha_j U}{f_{nj}} + \tau_{n_i,n_j,}\frac{N_{n_j}}{f_{n_j,}}\right). \tag{15}$$

- The execution time of subtask on cloud server

$$T_2(\alpha_d) = (t_2 + t_4) = \frac{\alpha_d U}{B_{n_i,d}} + \frac{\alpha_d U}{f_d} + \tau_{n_i,d}\frac{N_d}{f_d}. \tag{16}$$

- Overall execution time of aggregate task U

$$T(\alpha_j, \alpha_d) = \max\left\{\frac{\alpha_j U}{B_{n_i,n_j}} + \frac{\alpha_j U}{f_{n_j}} + \tau_{n_i,n_j}\frac{N_{n_j}}{f_{n_j}}, \frac{\alpha_d U}{B_{n_i,d}} + \frac{\alpha_d U}{f_d} + \tau_{n_i,d}\frac{N_d}{f_d}\right\}. \tag{17}$$

To minimalize the execution time, it is essential to define the optimum $\{\alpha_j \alpha_d\}$ set. Briefly, the problem is modeled by:

$$\min\left\{\max\left\{\frac{\alpha_j U}{B_{n_i,n_j,}} + \frac{\alpha_j U}{f_{nj}} + \tau_{n_i,n_j,}\frac{N_{n_j}}{f_{n_j,}}, \frac{\alpha_d U}{B_{n_i,d}} + \frac{\alpha_d U}{f_d} + \tau_{n_i,d}\frac{N_d}{f_d}\right\}\right\}. \tag{18}$$

$$\text{s.t.} \sum_{j=1}^{k}(\alpha_{jd} + \alpha) = 1$$

Here, the subtask allocated to every edge node fulfills $u_j = \alpha_j U$. Thus, the proportion of task assigned to targeted nodes and cloud form a $k1$ dimension vector $= (\alpha_1, \alpha_2, \ldots, \alpha_k, \alpha_{k+1})^{\hat{O}}$. Assume that the edge node n_1 obtains the existing task, the overall execution time T is defined by

$$T(\alpha) = \max\left(\frac{\alpha_1 U}{B_{n_1,n_1}} + \frac{\alpha_1 U}{f_{n_1}} + \tau_{n_1,n_1}\frac{N_{n_1}}{f_{n_1}}, \ldots, \frac{\alpha_k U}{B_{n_1,n_k}} + \frac{\alpha_k U}{f_{n_k}} + \tau_{n_1,n_k}\frac{N_{n_k}}{f_{n_k}}, \frac{\alpha_d U}{B_{n_1,d}} + \frac{\alpha_d U}{fd} + \tau_{n_1,d}\frac{N_d}{fd}\right). \tag{19}$$

As a result, the mapping of computation tasks is resolved in the case whereby the aggregated task U is named the proportion of task allocated to every targeted node, the solution of vector α. To prevent over-loading, a node afterward being allocated a massive number of tasks, we create the subtasks u lesser than or equivalent to the average load, that is, $u_j \le \frac{\left(U + \sum_{j=1}^{k} N_j\right)}{m}$. Assume that the overall amount of edge nodes is m, the problem is decreased to the subsequent optimization problem:

$$\alpha = \arg_{U \in I} \min\{T(\alpha)\} \tag{20}$$

$$\text{s.t. } 0 \le \alpha_j \le 1$$

$$0 \le \alpha_j \le 1$$

$$\alpha_j U \le \frac{\left(U + \sum_{j=1}^{k} N_j\right)}{m}$$

$$\sum_{j=1}^{k}(\alpha_{jd} + \alpha_d) = 1 \tag{21}$$

The searching space I for an optimization issue is given below and can be resolved by the use of ISROLBS-ECE technique.

$$I \underset{=}{\triangle} \prod_{j=1}^{k}[\alpha_{j\min}, \alpha_{j\max}] = \prod_{j=1}^{k}[0, 1]. \tag{22}$$

4 | PERFORMANCE VALIDATION

4.1 | Experimental setup

The simulation method is made up of a group of small VM bots with a lightweight Linux distribution (Ubuntu 16.04 LTS) and html5 web browsing capabilities. According to a report by the Natural Resources Défense Council (NRDC) in the United States data centers consumed 91 billion kWh (Kilo Watt Hours) of energy in 2013, the equivalent of 34 large coal-fired plants. The amount of energy consumed by Data centers are expected to reach 140 billion kWh by 2020 (53% increase), the equivalent annual output of 50 power plants, costing American businesses $13 billion in electricity bills each year and emitting nearly 100 million metric tonnes of carbon pollution. As the energy consumption of cloud computing systems has become one of the most critical issues for both the environment and cloud operational costs, many research efforts have been undertaken by both industry and academia to reduce the energy consumption of cloud underlying data centers.

These bots will ask the video service for online content. SRO equations will be used to grade each VM bot based on the initial buffering time, the number of stalls, the total stall time, and the average resolution quality of the content during each session. Environment parameters are Network size, node arrival, node capacity and simulation cycles. This section examines the performance of the ISROLBS-ECE model using CCTV video in edge computing environment. Figure 3 shows the sample CCTV sequence. The presented ISROLBS-ECE model is tested under varying number of mobile devices. The results are inspected in terms of different measures such as task waiting time (TWT), network traffic (NETT), average CPU usage of VMs (ACPUU-VM), and number of tasks successfully executed on edges (NTSEE), Power Consumption (PC), Memory Overhead (MO).

Table 1 and Figure 4 reports a detailed result analysis of the ISROLBS-ECE model with existing models[25–27] under 200 MDs. The results implied that the ISROLBS-ECE model has shown better performance with minimal values of TWT and NETT.[28,29] The ISROLBS-ECE model has attained

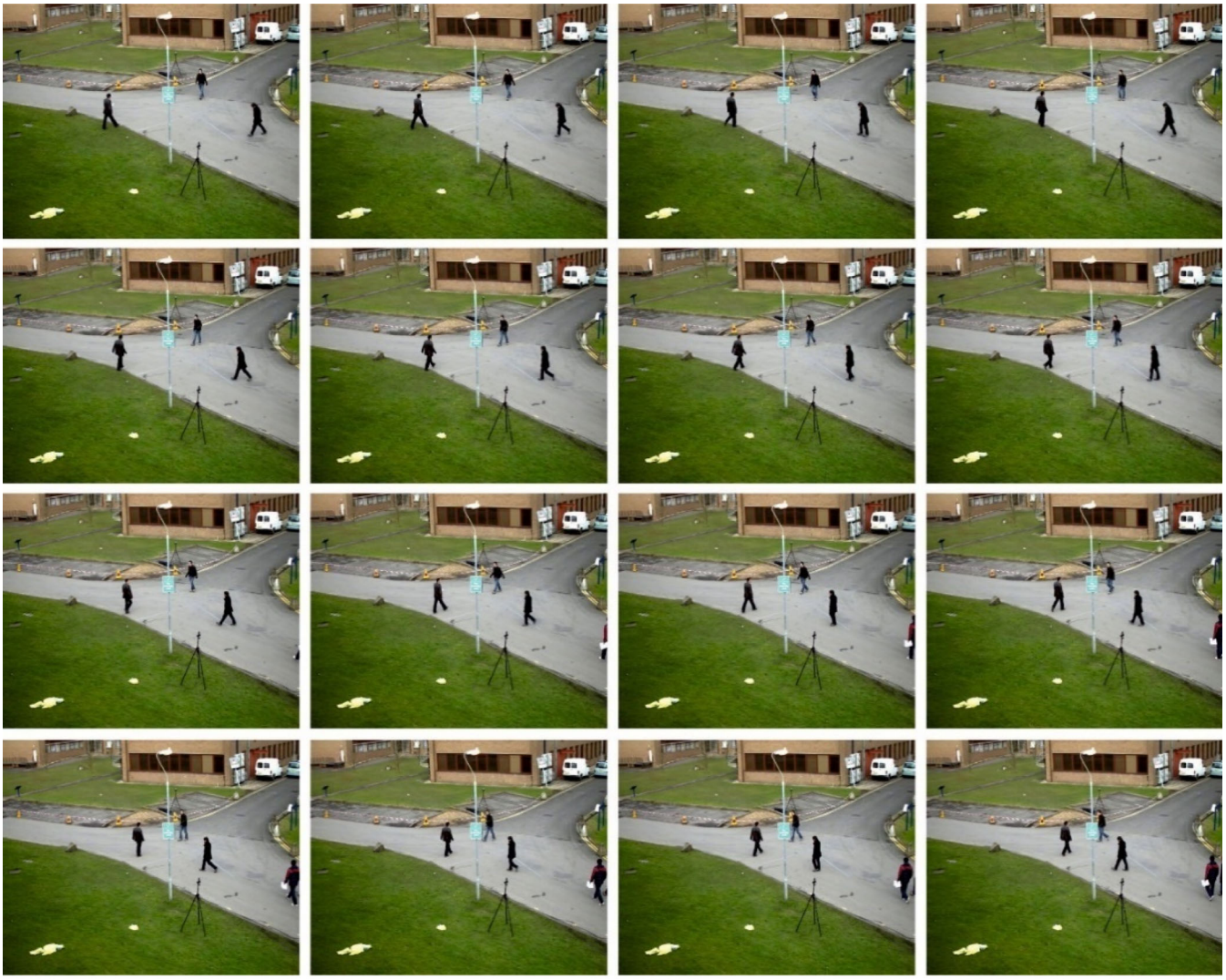


FIGURE 3 Sample CCTV sequences

TABLE 1 Load balancing results of ISROLBS-ECE model under 200 MDs

Number of MD = 200						
Methods	**TWT (s)**	**NETT (MB)**	**ACPUU-VM (%)**	**NTSEE**	**PC (W)**	**MO (%)**
Cloud First	0.220	1450	0.230	30	105.4	94.7
Round Robin	0.049	630	2.100	790	99.6	92.2
Priority	0.051	810	2.750	720	95.3	88.3
LBAMD-ECCE	0.030	600	2.900	870	90.7	87.5
ISROLBS-ECE	0.021	540	3.300	910	83.5	82.6

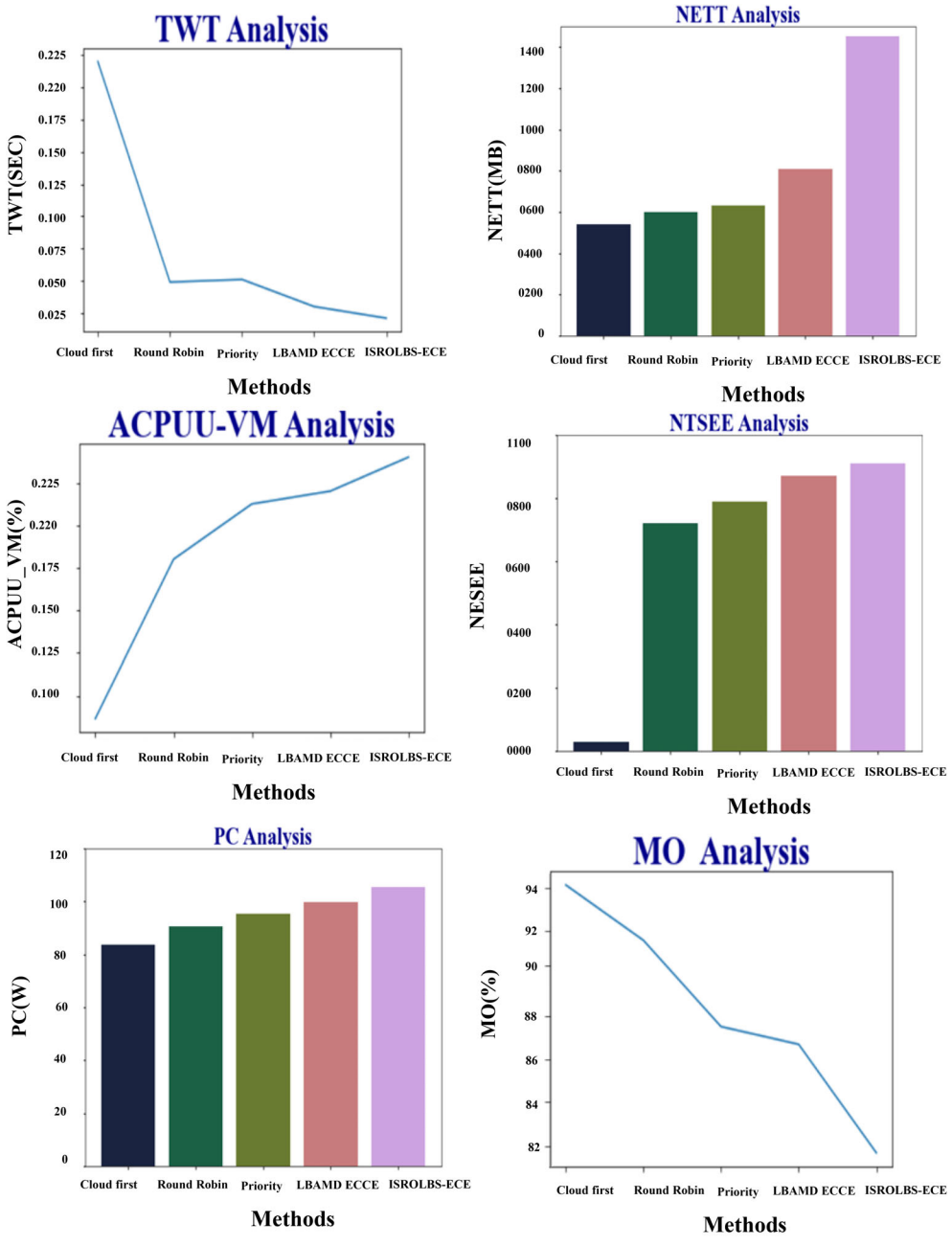


FIGURE 4 Comparative results of ISROLBS-ECE model on 200 MDs

maximum values of ACPUU-VM and NTSEE.[30,31] In addition, the results implied that the ISROLBS-ECE model has shown better performance with minimal values of PC and MO. Based on TWT, the ISROLBS-ECE model has offered lower TWT of 0.021 s whereas the CF, RR, Priority, and LBAMD-ECCE models have attained higher TWT of 0.220, 0.049, 0.051, and 0.030 s, respectively. Similarly, with respect to NTSEE, the ISROLBS-ECE model has provided improved NTSEE of 910 whereas the CF, RR, Priority, and LBAMD-ECCE models have achieved lower NTSEE of 30, 790, 720, and 870, respectively. Similarly, PC, the ISROLBS-ECE model has offered lower PC of 83.5 W whereas the CF, RR, Priority, and LBAMD-ECCE models have attained higher PC of 105.4, 99.6, 95.3, and 90.7 W, respectively. Similarly, MO, the ISROLBS-ECE model has offered lower MO of 82.6% whereas the CF, RR, Priority, and LBAMD-ECCE models have attained higher MO of 94.7%, 92.2%, 88.3%, and 87.5%, respectively.

Table 2 and Figure 5 shows a detailed analysis of the ISROLBS-ECE with current methods under 400 MDs. The result shows that the ISROLBS-ECE technique has shown improved performance with least values of TWT and NETT.

Likewise, the ISROLBS-ECE approach has accomplished higher value of ACPUU-VM and NTSEE. In addition, the results implied that the ISROLBS-ECE model has shown better performance with minimal values of PC and MO. Based on TWT, the ISROLBS-ECE technique has presented low TWT of 0.021 s while the CF, RR, Priority, and LBAMD-ECCE approaches have accomplished maximum TWT of 0.600, 0.100, 0.050, and 0.032 s, correspondingly. Likewise, regarding NTSEE, the ISROLBS-ECE technique has offered enhanced NTSEE of 1805 while the CF, RR, Priority, and LBAMD-ECCE systems have accomplished lower NTSEE of 200, 1650, 1700, and 1780, correspondingly. Similarly, PC, the ISROLBS-ECE model has offered lower PC of 130.5 W whereas the CF, RR, Priority, and LBAMD-ECCE models have attained higher PC of 205.4, 187.9, 172.8, and 153.8 W, respectively. Similarly, MO, the ISROLBS-ECE model has offered lower MO of 86.2% whereas the CF, RR, Priority, and LBAMD-ECCE models have attained higher MO of 96.1%, 94.2%, 91.4%, and 89.5%, respectively.

Table 3 and Figure 6 depicts a detailed analysis of the ISROLBS-ECE with current techniques under 600 MDs. The outcome shows that the ISROLBS-ECE approach has shown improved performance with least values of TWT and NETT. In addition, the ISROLBS-ECE technique has accomplished highest values of ACPUU-VM and NTSEE. In addition, the results implied that the ISROLBS-ECE model has shown better performance with minimal values of PC and MO. Based on TWT, the ISROLBS-ECE method has offered lower TWT of 0.272 s while the CF, RR, Priority, and LBAMD-ECCE approach have accomplished maximum TWT of 1.080, 0.520, 0.360, and 0.361 s, correspondingly. Likewise, regarding NTSEE, the ISROLBS-ECE system has presented enhanced NTSEE of 2613 while the CF, RR, Priority, and LBAMD-ECCE techniques have accomplished minimum NTSEE of 500, 2300, 2560, and 2490, correspondingly. Similarly, PC, the ISROLBS-ECE model has offered lower PC of 160.4 W whereas the CF, RR, Priority, and LBAMD-ECCE models have attained higher PC of 238.3, 218.7, 208.4, and 173.9 W, respectively. Similarly, MO, the ISROLBS-ECE model has offered lower MO of 87.4% whereas the CF, RR, Priority, and LBAMD-ECCE models have attained higher MO of 95.3%, 93.8%, 91.7%, and 89.3%, respectively.

Table 4 and Figure 7 demonstrates a detailed analysis of the ISROLBS-ECE with present methodologies under 800 MDs. The outcome illustrates that the ISROLBS-ECE approach has illustrated improve performance with least values of TWT and NETT. Likewise, the ISROLBS-ECE technique has accomplished maximal values of ACPUU-VM and NTSEE. In addition, the results implied that the ISROLBS-ECE model has shown better performance with minimal values of PC and MO. Based on TWT, the ISROLBS-ECE system has provided minimum TWT of 1.009 s while the CF, RR, Priority, and LBAMD-ECCE methodologies have achieved highest TWT of 1.810, 0.930, 1.174, and 1.135 s, correspondingly. Likewise, about NTSEE, the ISROLBS-ECE technique has offered enhanced NTSEE of 3484 while the CF, RR, Priority, and LBAMD-ECCE approaches have accomplished minimum NTSEE of 1380, 3080, 3210, and 3375, correspondingly. Similarly, PC, the ISROLBS-ECE model has offered lower PC of 190.3 W whereas the CF, RR, Priority, and LBAMD-ECCE models have attained higher PC of 288.3, 263.7, 243.6, and 222.5 W, respectively. Similarly, MO, the ISROLBS-ECE model has offered lower MO of 88.3% whereas the CF, RR, Priority, and LBAMD-ECCE models have attained higher MO of 96.4%, 94.7%, 92.8%, and 90.5%, respectively.

Lastly, Table 5 and Figure 8 illustrates a detailed analysis of the ISROLBS-ECE technique with current approaches under 1000 MDs. The result implies that the ISROLBS-ECE method has shown improved performance with least values of TWT and NETT. Similarly, the ISROLBS-ECE approach has reached maximal values of ACPUU-VM and NTSEE. In addition, the results implied that the ISROLBS-ECE model has shown better performance with minimal values of PC and MO.

Based on TWT, the ISROLBS-ECE method has provided minimum TWT of 1.690 s while the CF, RR, Priority, and LBAMD-ECCE techniques have obtained maximum TWT of 2.630, 1.920, 1.750, and 2.100 s, correspondingly. Likewise, about NTSEE, the ISROLBS-ECE method has offered better NTSEE of 4127 while the CF, RR, Priority, and LBAMD-ECCE techniques have attained minimum NTSEE of 2000, 3900, 3995, and 4000, correspondingly. By looking into the detailed results, it is confirmed that the ISROLBS-ECE technique has shown better load balancing performance over other models. Similarly, PC, the ISROLBS-ECE model has offered lower PC of 220.3 W whereas the CF, RR, Priority, and LBAMD-ECCE models have attained higher PC of 324.4, 290.7, 267.3, and 246.9 W, respectively. Similarly, MO, the ISROLBS-ECE model has offered lower MO of 89.4% whereas the CF, RR, Priority, and LBAMD-ECCE models have attained higher MO of 98.6%, 96.3%, 94.8%, and 91.7%, respectively.

TABLE 2 Load balancing results of ISROLBS-ECE model under 400 MDs

Number of MD = 400						
Methods	**TWT (s)**	**NETT (MB)**	**ACPUU-VM (%)**	**NTSEE**	**PC (W)**	**MO (%)**
Cloud First	0.600	2950	0.300	200	205.4	96.1
Round Robin	0.100	1500	2.200	1650	187.9	94.2
Priority	0.050	1750	2.410	1700	172.8	91.4
LBAMD-ECCE	0.032	1480	2.750	1780	153.8	89.5
ISROLBS-ECE	0.021	1395	3.000	1805	130.5	86.2

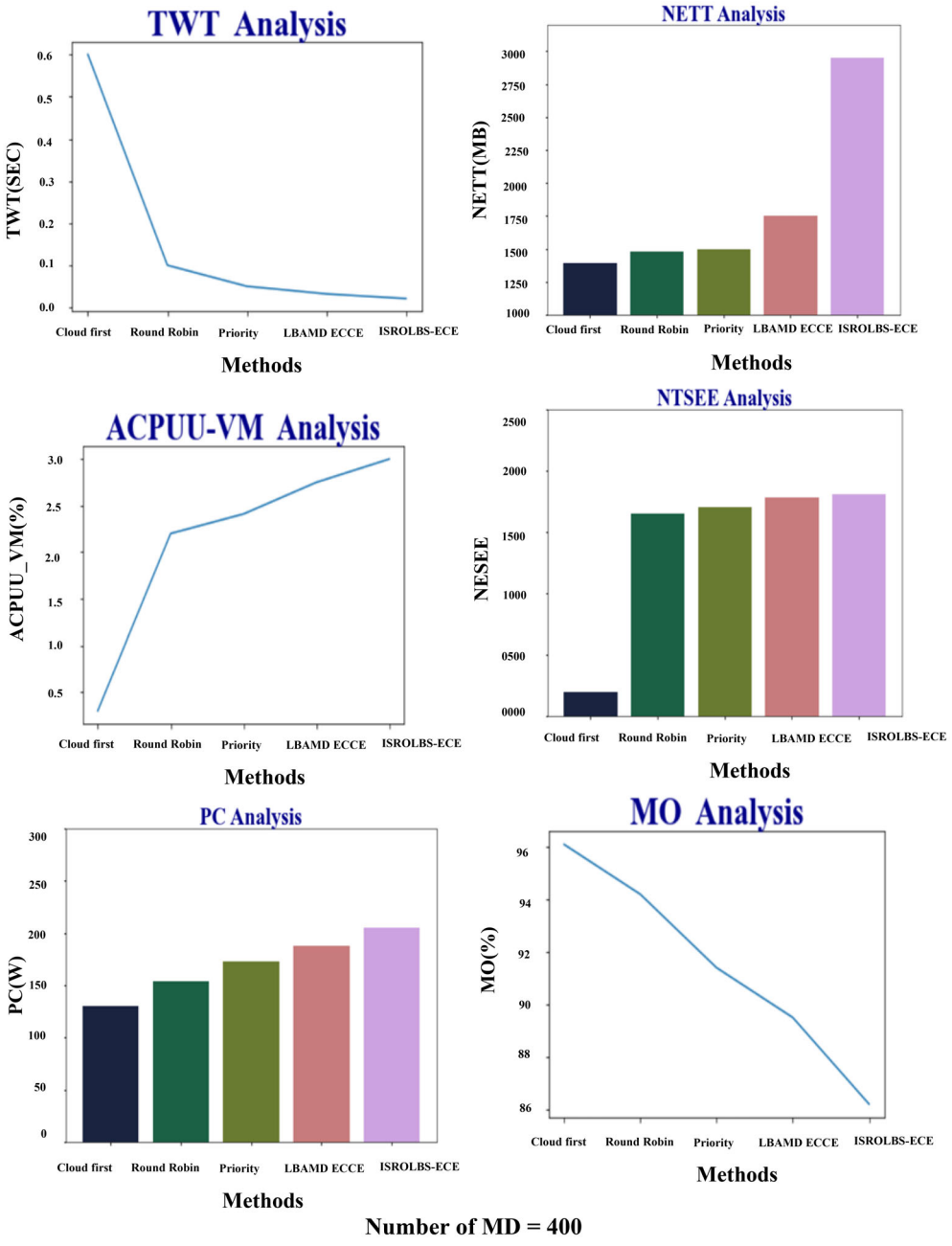


FIGURE 5 Comparative results of ISROLBS-ECE model on 400 MDs

TABLE 3 Load balancing results of ISROLBS-ECE model under 600 MDs

Number of MD = 600						
Methods	**TWT (s)**	**NETT (MB)**	**ACPUU-VM (%)**	**NTSEE**	**PC (W)**	**MO (%)**
Cloud First	1.080	3490	0.330	500	238.3	95.3
Round Robin	0.520	2260	1.610	2300	218.7	93.8
Priority	0.360	2500	2.130	2560	208.4	91.7
LBAMD-ECCE	0.361	2510	3.000	2490	173.9	89.3
ISROLBS-ECE	0.272	2109	3.265	2613	160.4	87.4

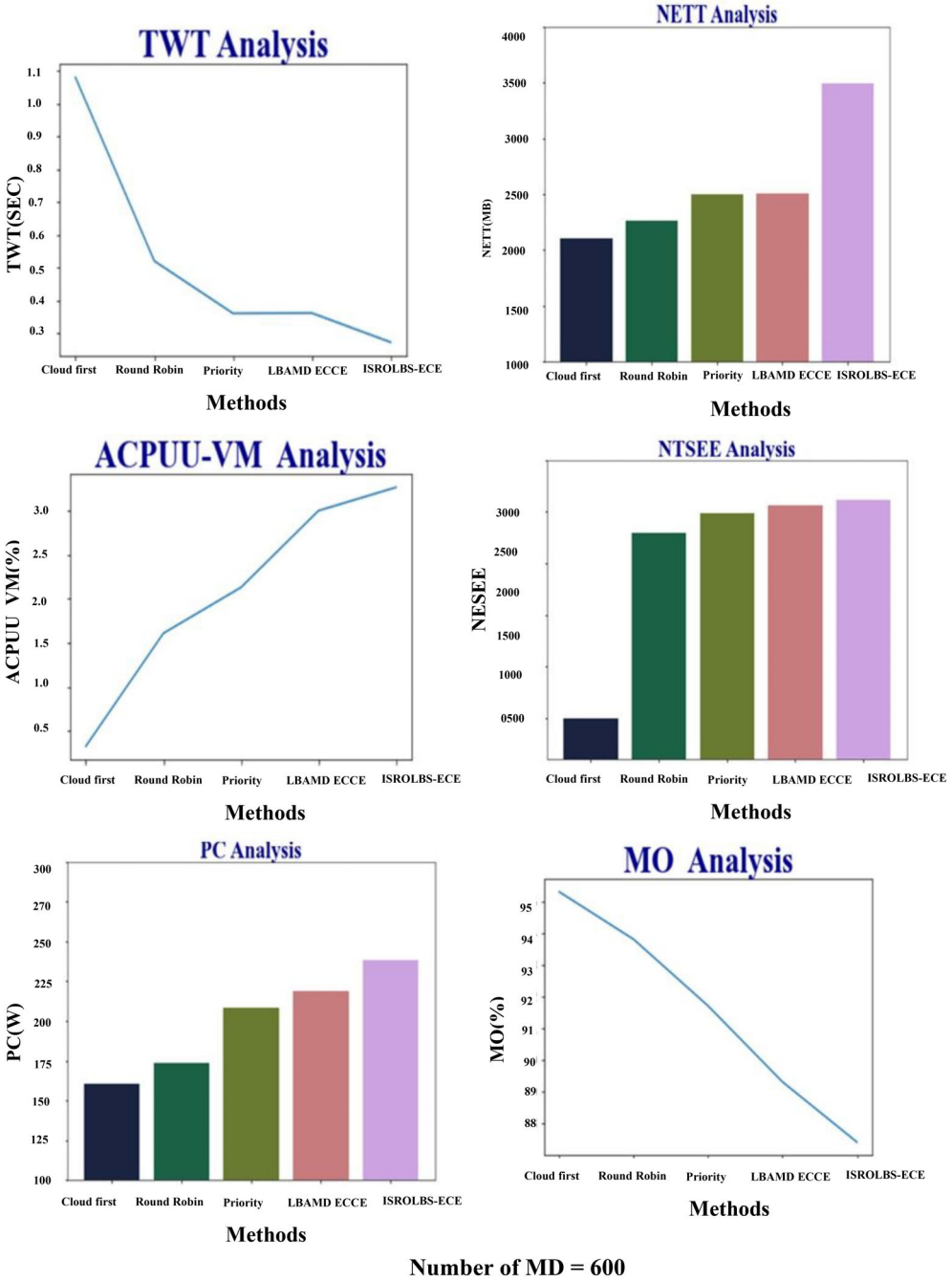


FIGURE 6 Comparative results of ISROLBS-ECE model on 600 MDs

TABLE 4 Load balancing results of ISROLBS-ECE model under 800 MDs

Number of MD = 800						
Methods	**TWT (s)**	**NETT (MB)**	**ACPUU-VM (%)**	**NTSEE**	**PC (W)**	**MO (%)**
Cloud First	1.810	4950	0.510	1380	288.3	96.4
Round Robin	0.930	3600	1.490	3080	263.7	94.7
Priority	1.174	3800	2.389	3210	243.6	92.8
LBAMD-ECCE	1.135	3200	2.768	3375	222.5	90.5
ISROLBS-ECE	1.009	2860	2.935	3484	190.3	88.3

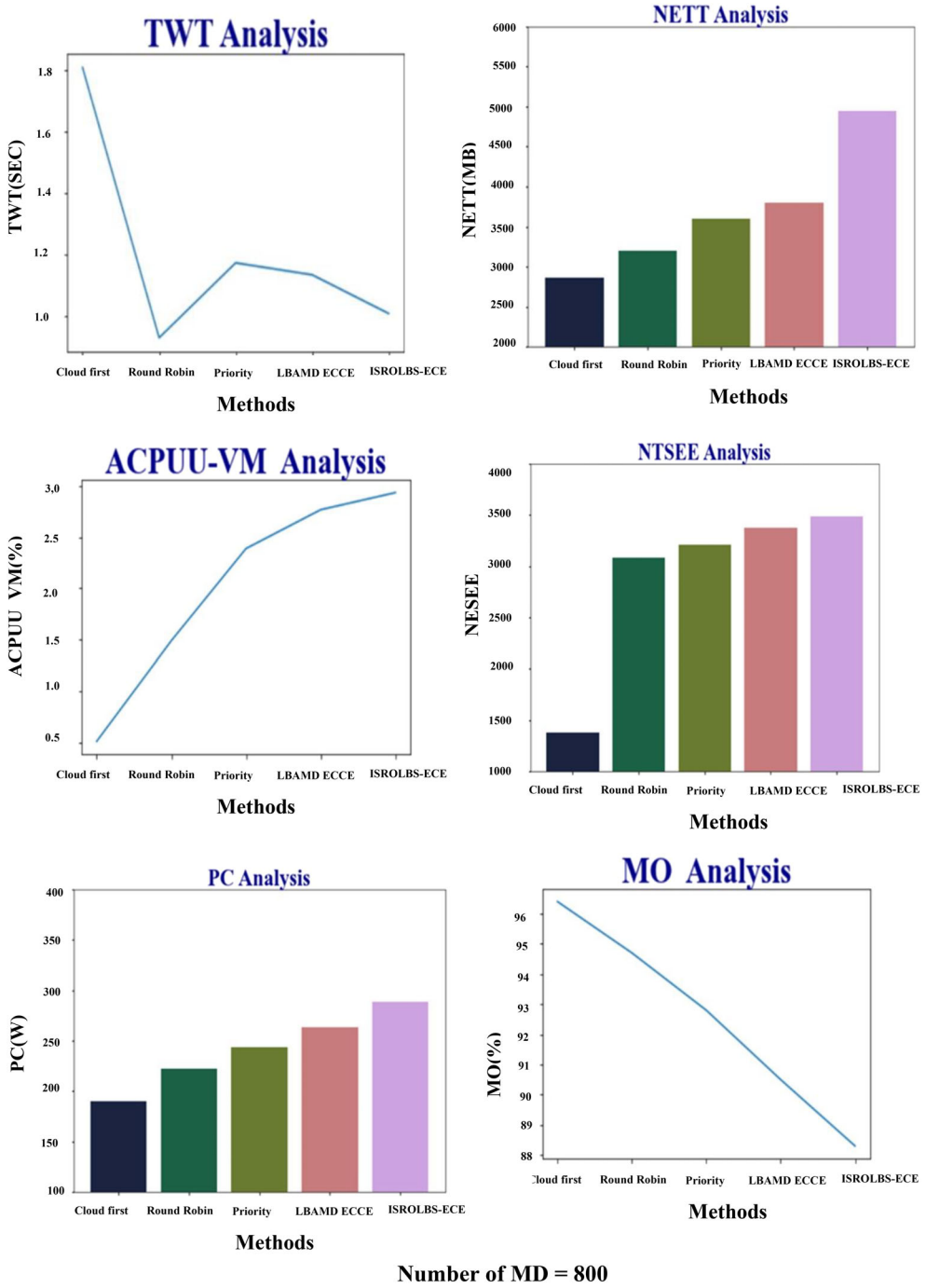


FIGURE 7 Comparative results of ISROLBS-ECE model on 800 MDs

TABLE 5 Load balancing results of ISROLBS-ECE model under 1000 MDs

Number of MD = 1000						
Methods	**TWT (s)**	**NETT (MB)**	**ACPUU-VM (%)**	**NTSEE**	**PC (W)**	**MO (%)**
Cloud First	2.630	5500	1.250	2000	324.4	98.6
Round Robin	1.920	4800	1.620	3900	290.7	96.3
Priority	1.750	4900	2.365	3995	267.3	94.8
LBAMD-ECCE	2.100	4200	2.972	4000	246.9	91.7
ISROLBS-ECE	1.690	4125	3.212	4127	220.3	89.4

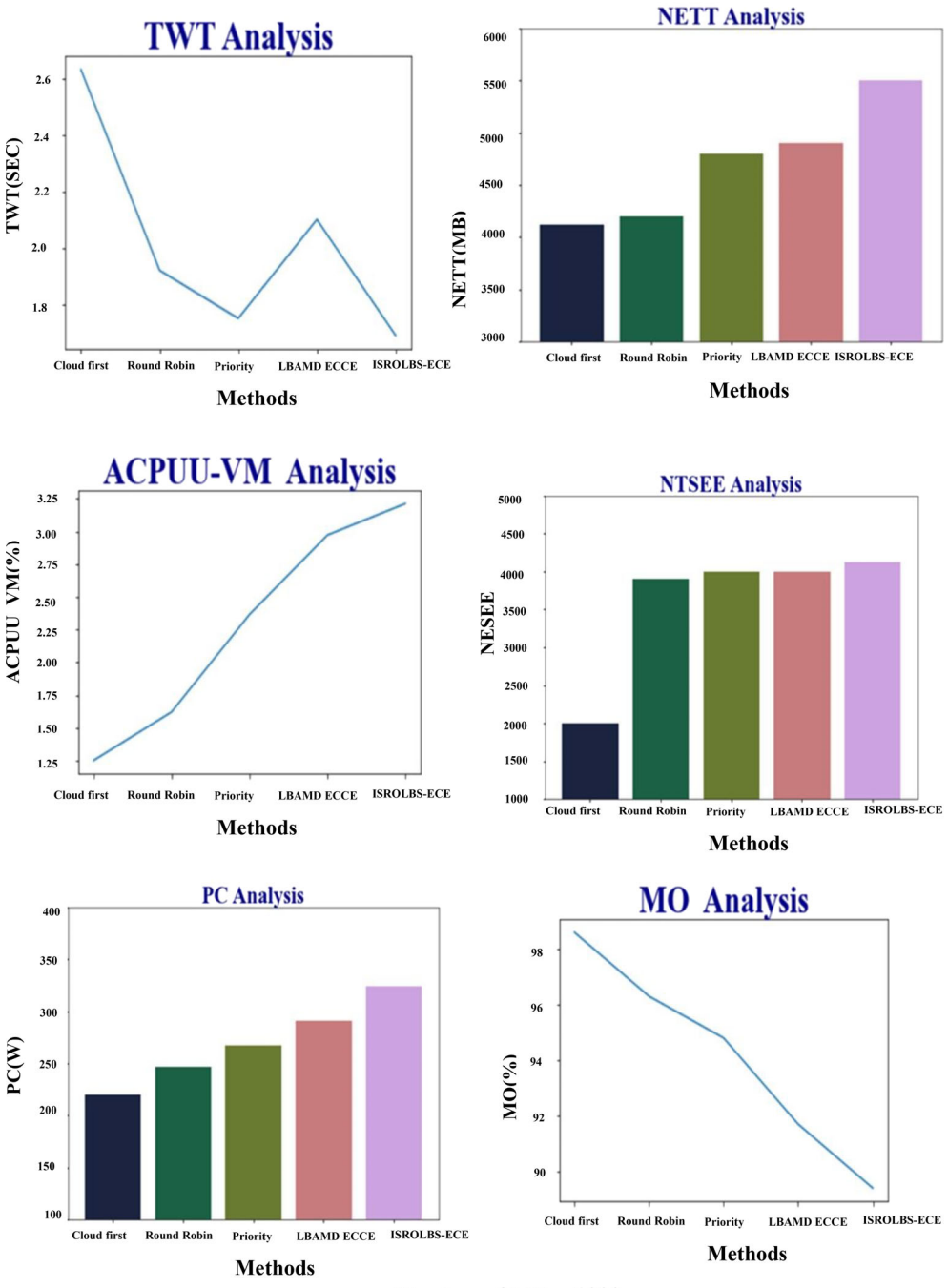


FIGURE 8 Comparative results of ISROLBS-ECE model on 1000 MDs

5 | CONCLUSION

In this study, a new ISROLBS-ECE methodology has been developed for distributing the offloaded tasks to neighboring edge servers effectively. Besides, the ISROLBS-ECE technique involves the hybridization of chaotic concepts into the conventional SRO algorithm. In this paper, we consider solving the problem of task offloading and resource allocation in the multi-mobile UE and multi-server MEC environment. A new adaptive algorithm is proposed to reduce the average task response time and the energy consumption of the entire system. We model the objective problem as an MDP, select the appropriate computing node for each task based on the intelligent learning ability of the ISROLBS-ECE method, and complete the task offloading and resource allocation process. In the ISROLBS-ECE technique, the comparatively lightest and task arrival nodes are employed as target nodes for allocating new tasks whereas the other nodes are not provisionally assigned tasks, to accomplish dynamic balancing of the network. The performance evaluation of the ISROLBS-ECE technique take place and the outcomes are investigated under different aspects. The comparative study shows the better performance of the ISROLBS-ECE approach over other recent techniques. Therefore, the ISROLBS-ECE algorithm can be applied as an effectual tool for load balancing in the edge computing environment. In the future, the projected ISROLBS-ECE methodology can be extended to the design of encryption systems to accomplish privacy and security. In addition, Throughput and computational capacity-based scaling techniques will be introduced in cloud hosting costs and cooling down durations. In generally lack the agility to comprehend Quality of Experience (QoE) degradation. In future, to bring forward a solution for these circumstances, a QoE scaling technique will be used to considers all aspects of on-line video delivery that shows outstanding performance when compared with conventional cloud scaling strategies.

CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest to report regarding the present study

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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